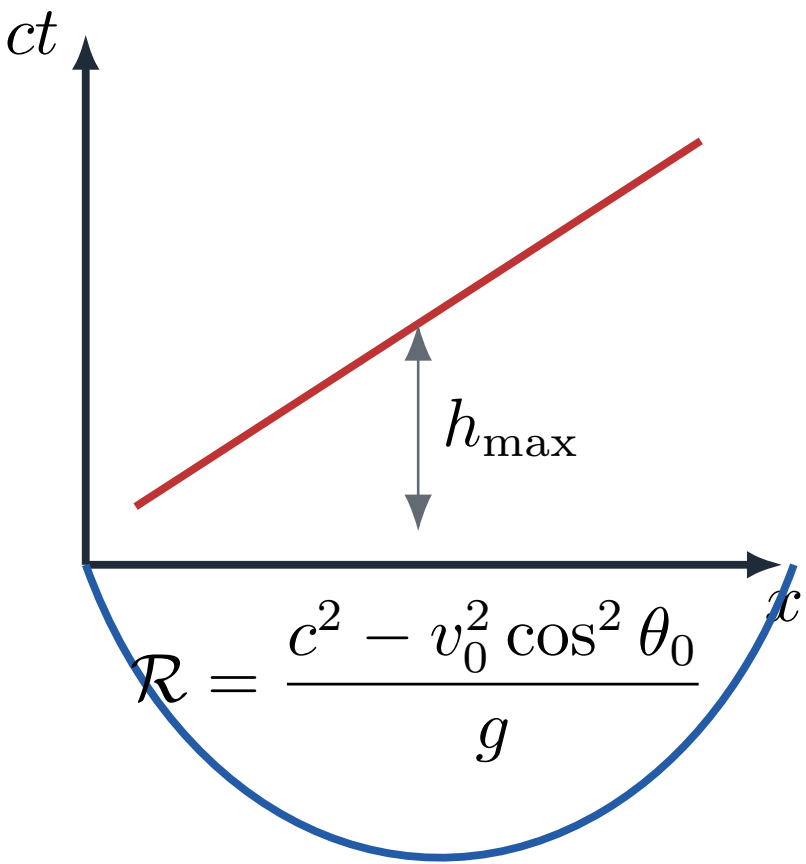
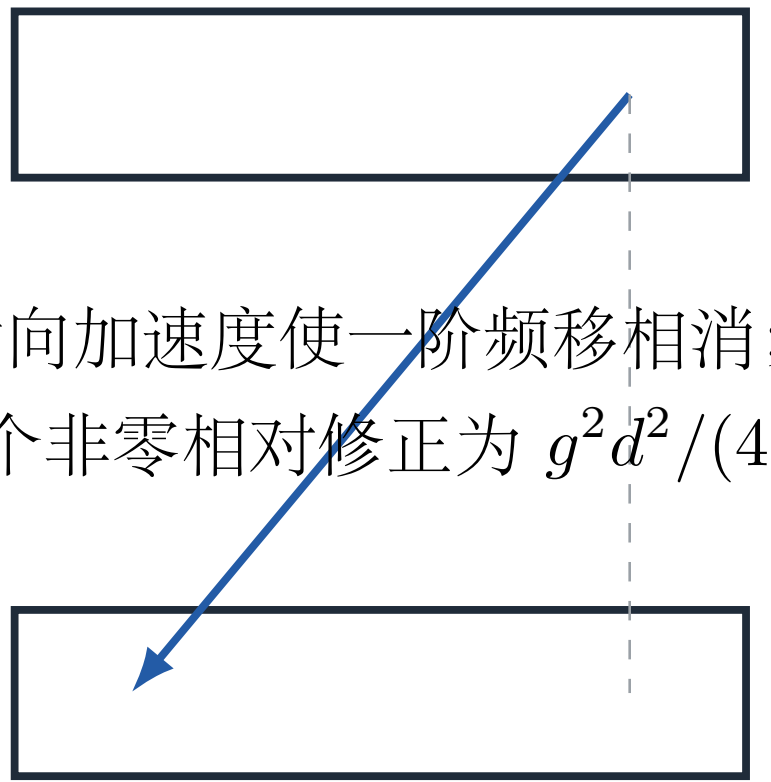




横向加速度使一阶频移相消；

首个非零相对修正为 $g^2 d^2/(4c^4)$